Course Code : ECT 257 **MQNR/RS – 23 / 1181**

Fourth Semester B. Tech. (Electronics and Communication Engineering) Examination

ANALOG CIRCUITS

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions carry marks as indicated.

(2) Assume suitable data wherever necessary.

1. Consider the circuit shown in Fig. 01.

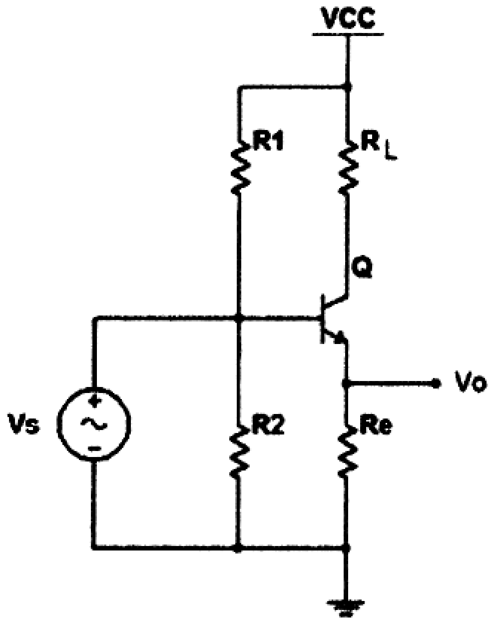

Fig. 1

(a) Identify the feedback topology used in the amplifier. Justify. 2(CO1)

(b) Analyze the amplifier to calculate equations for voltage gain, input resistance and output resistance of the amplifier circuit. 8(CO2)

2. (a) Elaborate mathematically the effect of load coupling in class A power amplifier. 8(CO3)

(b) Explain cross-over distortion. 2(CO1)

3. (a) Explain working of Hartley oscillator. 6(CO2,3)

(b) Determine the oscillation frequency and feedback fraction for a transistorized Hartley oscillator having $L_1 = L_2 = 100\ \mu H$ and $C = 20$ pF. 4(CO2,3)

4. (a) Derive gain, input resistance and output resistance of a dual input balanced output differential amplifier. 8(CO3)

(b) Explain the use of swamping resistor in differential amplifier. 2(CO3)

5. (a) Design a Schmitt trigger using OP-AMP to obtain threshold voltage V_{ut} and V_{lt} as $+1$ V and $-1{\cdot}4$ V respectively. Assume supply voltage of ± 12 V and input of 1 kHz, 4 Vpp. Sketch input and output waveforms. 6(CO4)

(b) Design a circuit using OP-AMP to implement following input output relation :

$$Vo = -V_a - 2V_b - 2{\cdot}5V_c$$ 4(CO4)

6. (a) Design a square wave generator using IC 555 having an output frequency of 1 kHz. 5(CO5)

(b) Design a second order LPF using OP-AMP with high cut-off frequency of 2 kHz. 5(CO5)